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N-MON - Team G

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GNSS Jamming Monitor

User Manual

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1 Introduction

1.1 What is N-MON

N-MON (Navigation Monitor) is a compact, autonomous device designed to continuously monitor GNSS (Global Navigation Satellite System) signals on vehicles, such as trucks transporting high-value cargo, and immediately alert fleet operators when a jamming attack is detected.

GNSS systems allow fleet management companies to track their vehicles in real time. Unfortunately, cheap devices called jammers can deliberately disrupt these signals, causing a vehicle to silently disappear from the monitoring system. This vulnerability is actively exploited by criminal organisations involved in cargo theft. In Europe alone, the estimated annual losses exceed €8.2 billion, and a single stolen truck can represent a loss of hundreds of thousands of euros.

N-MON was designed to fill the gap between two inadequate extremes: basic power-spike detectors that generate too many false alarms, and military grade anti-jamming systems that cost tens of thousands of euros per unit and are completely impractical for a commercial fleet. Instead, N-MON offers an intelligent, affordable, and fully autonomous solution that can be deployed on any truck. It does four things:

- it continuously listens to the GNSS frequency band (L1, 1575.42 MHz)
- it uses Artificial Intelligence to distinguish between a clean signal and a jamming attack
- it classifies the type of interference detected
- it immediately sends an alert to both the driver and the central control station

1.2 How N-MON works

Once powered on, N-MON runs its analysis pipeline continuously and automatically, without any input from the driver. The process repeats every 100 microseconds and consists of four stages:

1. **Signal capture:** A GNSS patch antenna mounted on the vehicle roof picks up the radio signals in the L1 band. An SDR (Software-Defined Radio) front end digitises these signals at 10 million samples per second, turning the analogue radio wave into a stream of digits that the onboard computational unit can process
2. **Analysis:** The onboard computer takes each 100-microsecond snapshot of raw signal data and converts it into a spectrogram, a two-dimensional image showing how the radio energy is distributed across different frequencies over time. This representation is particularly effective at capturing the distinctive patterns that jamming signals have
3. **Classification:** A lightweight AI model, specifically a Convolutional Neural Network (CNN) neural network, analyzes each spectrogram. Because jamming signals have characteristic patterns in time and frequency, it can identify jamming even at low power levels, before the vehicle GPS receiver loses its position fix.

4. **Alerting:** If jamming is detected, N-MON immediately publishes an alert through a secure internet connection to both the driver and the Central Control Centre

NOTE N-MON is a monitoring and alerting system. It does not block or counteract jamming signals. Its role is to detect and report them so that the appropriate response can be taken by the fleet operator.

2 System Overview

2.1 Hardware Components

The N-MON device consists of the following physical components:

Component	Model	Purpose
GNSS patch antenna	GPS P1MAM	Receives GNSS band radio signals from the vehicle exterior
SDR front-end	HackRF One	Digitises the analogue RF signal captured by the antenna
Processing Unit	NVIDIA Jetson Orin Nano	Runs the signal processing pipeline and the AI model in real time
Receiver Module	ZED-F9P module	Provides the vehicle's GPS position, which is attached to every jamming alert sent to the control centre

2.2 Software Architecture

The software on N-MON is structured as a pipeline of four sequential blocks. Each block hands its output to the next:

- **Signal acquisition:** GNU Radio reads IQ samples from the HackRF front-end at 10 MHz and feeds them into the processing pipeline
- **Data processing:** Short-Time Fourier Transform (STFT) converts each 100-microsecond snapshot into a spectrogram matrix, while in parallel 16 features are extracted from a 200-microsecond snapshot to achieve higher model accuracy
- **AI inference:** The CNN model classifies the spectrogram into one of six categories: Clean signal, or one of five jamming types (Linear Wideband Fast, Linear Narrowband, Triangular, Triangular Wave, Tick) and relative jamming power (Low, Mid, High)
- **Communication:** the classification result, together with the current GPS position from the Receiver Module, is published via MQTT to the cloud broker, which forwards it to the Control Centre and Driver Interfaces in real time

2.3 Communication Architecture

N-MON uses the MQTT protocol to send alerts. MQTT was specifically designed for devices that need to remain connected in environments with intermittent or low-bandwidth network coverage, which makes it a natural fit for trucks moving through tunnels, rural roads, and underground loading docks.

Key properties of the communication layer:

- **Reliable delivery:** alerts are sent with Quality of Service level 1 (QoS 1), which means MQTT will keep retrying until the message is confirmed as received by the broker even if the truck briefly loses signal
- **Offline detection:** N-MON registers a Last Will and Testament (LWT) with the broker when it starts up. If the device loses power or connectivity unexpectedly, the broker automatically notifies the control centre with an OFFLINE message associated to that vehicle's plate.
- **Security:** all messages are encrypted over TLS and require username/password authentication. Only authorised clients can subscribe to a vehicle's alert topics.

Each vehicle is identified by a unique *vehicle_id* (its licence plate), which appears in all MQTT topics:

- NMON/alert/*vehicle_id* : jamming detection alerts
- NMON/status/*vehicle_id* : periodic keep-alive messages

Alert messages are published only after 3 consecutive signal snapshots are classified with the same jamming type. This design choice was deliberately introduced to reduce false positives: requiring consistency across multiple snapshots ensures that a single anomalous reading does not trigger an unnecessary alarm, avoiding both the driver and the Control Centre operator to respond to events that do not represent a real threat.

In addition to alert messages, N-MON periodically publishes a status message every 10 seconds. This keep-alive mechanism allows both the Driver Interface and the Control Centre dashboard to continuously verify that the device is powered on and connected to the network. If no status message is received within a configured timeout period, the MQTT broker automatically publishes a Last Will and Testament message on behalf of the device, and the vehicle is marked as OFFLINE in the Control Centre dashboard and Driver Interface, signalling a potential loss of power or connectivity.

3 Installation

3.1 Antenna Placement

The GNSS patch antenna must be mounted on the exterior roof of the vehicle to ensure an unobstructed view of the sky. Avoid placing it near metal structures that could block or reflect satellite signals.

- Use a magnetic base or adhesive mount to fix it to the vehicle roof
- Route the coaxial cable avoiding sharp bends that could damage the cable and connect it to the SMA antenna port located on the top panel of the N-MON unit

The antenna signal is split internally via a splitter: one output feeds the SDR front-end for jamming detection, and the other feeds the ZED-F9P GPS receiver module for position reporting. This splitter is pre-wired inside the N-MON unit and requires no action from the installer.

3.2 Network Configuration

N-MON requires an internet connection to publish jamming alerts to the cloud broker. Two connectivity options have been tested and verified.

Option A: Ethernet

Connect the device to a router or network switch using a standard Ethernet cable (RJ45). The device will obtain network parameters automatically and establish the connection to the broker at every boot with no further configuration required

Option B: WiFi

Connect the device to a Wi-Fi network with internet access using the system network settings. To do this, connect a mouse, keyboard, and a DisplayPort cable attached to a monitor to the device. Once configured, the device will automatically reconnect to the network at every boot.

3.3 Power Supply

N-MON is powered using the supplied DC power adapter, which can be connected to any standard 220V AC outlet. Once connected, a green light on the Computational Unit turns on together with the blue light on the Receiver Module. If the lights do not turn on, it means that the system has lost power and the LWT was triggered due to a power failure rather than a connectivity issue. Once powered on, the device starts operating automatically, with no further action required.

4 Using the System

4.1 Driver Mobile Interface

The driver interface is a lightweight web page that can be opened on any smart-phone or tablet, no app needs to be installed. It is designed to be as simple as possible: the driver only needs to see whether the GNSS signal is clean or compromised. To open the driver interface:

- Ask the fleet manager for the N-MON portal URL or use the provided QR code at the end of the manual
- Open the URL in a mobile browser
- Type the vehicle's licence plate in the input field (e.g. AB123CD) and tap Connect & Monitor

The interface moves through the following states as shown in Figure 1:

- **Startup** (1a): the page loads and shows the licence plate input field
- **Offline** (1b): the plate has been entered but the device is not sending data. This typically means N-MON is powered off or has lost internet connectivity
- **Online** (1c): the device is active and monitoring. The system status indicator turns green.
- **Jamming detected** (1d): a jamming alert has been received. A red warning banner appears with the text JAMMING DETECTED. After the Acknowledge button is pressed there is a cooldown period of 30 seconds

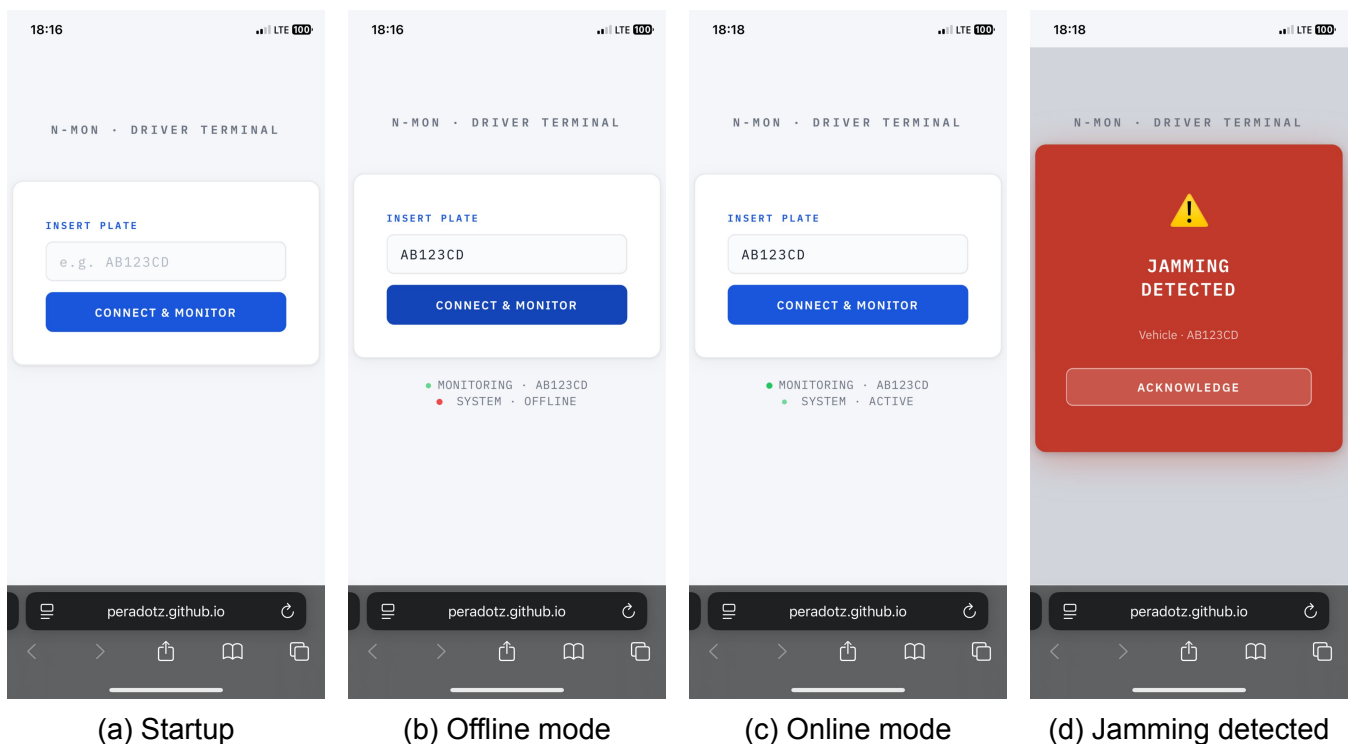


Figure 1: Driver Interface states and alerts

NOTE The driver interface is read-only. The driver does not need to configure anything or interact with the device. The only action required is to acknowledge the alert and then follow the company emergency protocol.

4.2 Control Centre Interface

The control centre dashboard is a web application built with Streamlit, accessible from any desktop browser using the credentials provided by the system administrator. It is designed for fleet operators who need a real-time overview of all monitored vehicles and a detailed record of every jamming event.

At the top of the dashboard, a summary panel shows three key statistics: the number of currently active vehicles, the total count of jamming events detected since the session started, and the system status (OPERATIONAL or DISCONNECTED from the broker).

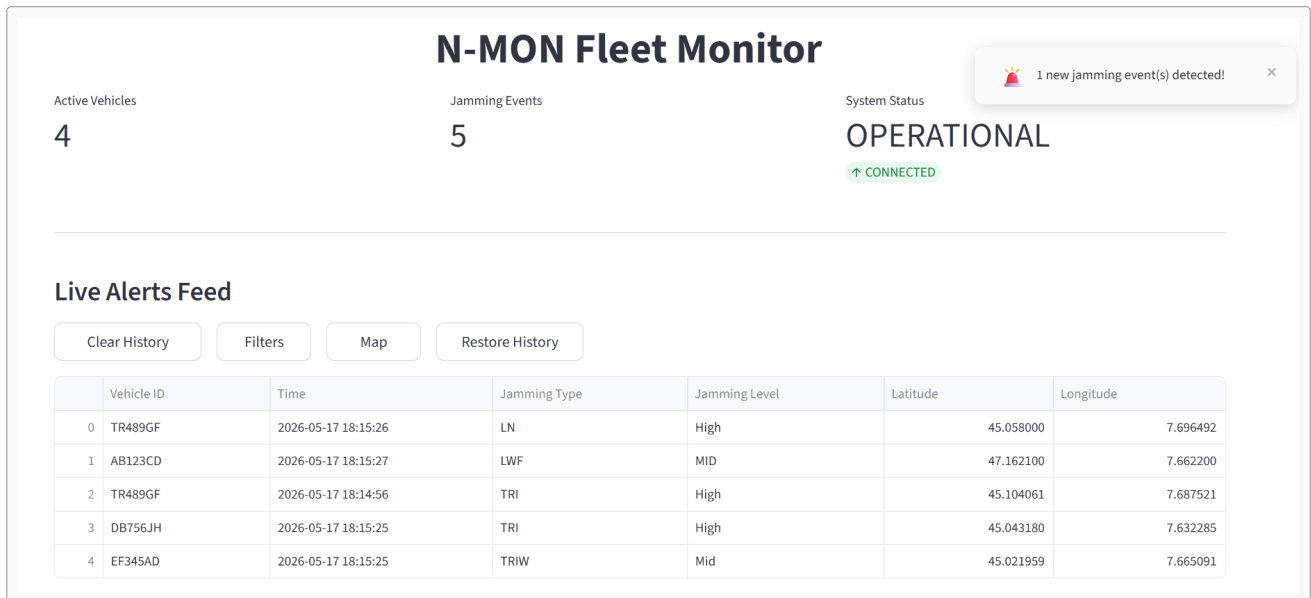


Figure 2: Live Alerts Feed with real-time notification banner

Below the summary panel, the Live Alerts Feed table (Fig. 2) lists every jamming event received. For each event, the table shows the vehicle plate, the jamming type, the intensity level, and the GPS coordinates at the time of detection. When a new alert arrives, a notification banner appears in the top-right corner of the screen and the jamming events counter updates immediately.

Operators can interact with the feed using four buttons:

- **Clear History:** removes all entries in the Alert Table from the current session log and saves them in a external csv file
- **Filters:** opens a panel to filter the alert table by vehicle, jamming type, intensity level or time. The same filters are automatically applied to the map view
- **Map:** switches the view to an interactive map showing the geographic position of each alert, which is useful for identifying patterns or recurring problem areas on a specific route
- **Restore History:** restores all the entries in the Alert Table that have been saved in the csv file

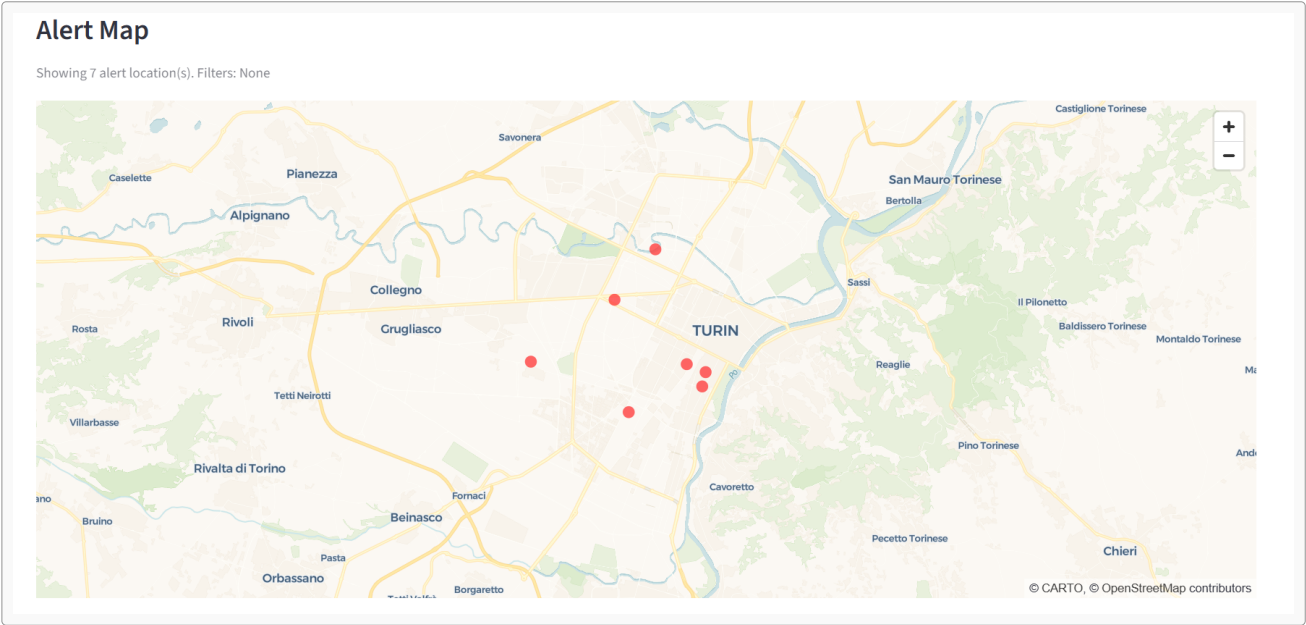


Figure 3: Map Visualization

At the bottom of the dashboard, the Active Vehicles Table (Fig. 4) shows all vehicles currently connected to the system. For each vehicle, the table displays the plate number, the current connection status (ACTIVE or OFFLINE), the timestamp of the last received message, and the last known GPS position. A vehicle is marked OFFLINE automatically by the broker within 60 seconds of an unexpected disconnection, via the Last Will and Testament mechanism configured at startup. If it remains offline for an extended period, it is eventually removed from the Active Table.

1	TR489GF	2026-05-17 18:15:55	LN	High	45.043514	7.713335
2	AB123CD	2026-05-17 18:16:03	LWF	MID	48.062100	7.662200
3	TR489GF	2026-05-17 18:14:56	TRI	High	45.104061	7.687521
4	DB756JH	2026-05-17 18:15:25	TRI	High	45.043180	7.632285
5	EF345AD	2026-05-17 18:15:55	TRIW	Mid	45.032381	7.681368

See Active Vehicles

	Vehicle ID	Status	Last Seen	Latitude	Longitude
0	EF345AD	ACTIVE	2026-05-17 18:16:15	45.016465	7.667860
1	TR489GF	ACTIVE	2026-05-17 18:16:19	45.035690	7.709035
2	DB756JH	ACTIVE	2026-05-17 18:16:19	45.011601	7.648713
3	AB123CD	OFFLINE	2026-05-17 18:16:07	48.162100	7.662200

Figure 4: Active Vehicles Table

5 Technical Appendix

5.1 ResNet-18 Architecture and Feature Fusion

The classification backbone is ResNet-18 (a Residual Network with 18 layers and around 11.7 million parameters), chosen for its balance between learning capacity and computational efficiency on the Nvidia Jetson Orin Nano. The model was initialised with ImageNet pre-trained weights to speed up convergence.

5.1.1 Architecture Modifications

The original architecture was adapted as follows:

1. **Input Layer:** The first convolutional layer was modified from 3-channel RGB to single-channel int8 spectrograms produced via Short-Time Fourier Transform (STFT).
2. **Feature Extraction:** The ResNet backbone outputs 512-dimensional feature vectors.
3. **Feature Fusion:** 16 hand-crafted statistical features (mean, median, std, RMS, percentiles, skewness, kurtosis, entropy) are batch-normalised and concatenated with the ResNet features, giving a 528-dimensional vector.
4. **Classification Head:** $528 \rightarrow 256$ (ReLU, Dropout 0.5) $\rightarrow$ 16 classes. An energy score (log-sum-exp of logits) is computed per sample for open-set detection.

5.1.2 Dataset and Training

The dataset contains 44,428 samples across 16 classes: CLEAN + 5 jammer types $\times$ 3 power levels (LOW, MID, HIGH) based on Jammer-to-Signal Ratio. Data was split 50/15/35% (train/val/test, stratified). Training used Adam (lr=0.001), cross-entropy loss, batch size 128, and early stopping (patience=5). Convergence typically occurred within 20 epochs on GPU.

5.2 In-Set Classification and Open-Set Recognition

5.2.1 Closed-Set Baseline

Without open-set recognition, the model always forces inputs into one of the 16 known classes via softmax. On known samples it performs well, reaching an overall accuracy of 95%. However, it completely fails on unknown jammer types: all unknown samples in the extended validation set are assigned to a known class. The failure is not random, it is highly systematic, most of all unknowns are misclassified as TICK_LOW, regardless of their actual characteristics. This reveals that unknown signal embeddings consistently fall close to the TICK_LOW centroid in the 528-dimensional feature space. When tested on a dataset containing only unknown jamming types, the model still detects the presence of jamming in 78% of cases. Even though the specific jammer type is misclassified, the system remains reliable at alerting users that a jamming attack is occurring. However, this is a critical problem for deployment: encountering a new jammer type produces high

rates of misclassification. Softmax scores are not suitable for unknown detection, but the model's internal feature space preserves enough geometric structure for distance-based methods to work — which is exactly what OSR exploits.

5.2.2 Open-Set Recognition Framework

OSR operates on the 528-dimensional penultimate feature space and enables both unknown detection and error correction via distance-based decision boundaries.

Implementation:

1. **Centroid Computation:** Class centroids are computed by averaging training embeddings per class.
2. **Distance Metric:** L2 distance from each sample's embedding to all class centroids is computed at inference time.
3. **ROC-based Threshold Calibration:** A per-class threshold is set by capping the False Positive Rate (unknowns incorrectly accepted) at $\leq 5\%$, while maximising the True Positive Rate (known samples correctly accepted).
4. **Binary Decision Logic:**

ACCEPT : Distance to predicted class $\leq$ threshold $\rightarrow$ keep prediction

REJECT : Predicted class is not within threshold $\rightarrow$ classify as UNKNOWN

5.2.3 Unknown-Only Dataset Evaluation

To isolate OSR performance only on new signals, we constructed a test set containing only unknown jammer types: Frequency-Hopping (FH), Hooked-Sawtooth (HS), Linear-Wide-Slow (LWS), and Multitone-Narrow (MN). Each type had 600 samples from index ranges outside the training distribution, ensuring no overlap with the 16 known classes (same dataset used in the previous section).

OSR detection performance varies significantly across jammer types. The overall unknown detection rate is **32.8%** (786/2400), with strong variation:

- **MULTITONE-NARROW:** 57.3% detected as unknown — best performance
- **FREQUENCY-HOPPING:** 45.7% detected as unknown
- **HOOKEED-SAWTOOTH:** 15.2% detected as unknown
- **LINEAR WIDE SLOW:** 12.8% detected as unknown — worst performance

The confusion matrix reveals systematic misclassification patterns. A significant portion of unknown samples are classified as CLEAN (38.3% for FH, 39.0% for HS, 29.2% for LWS, 38.5% for MN), suggesting these jammer embeddings fall near the CLEAN centroid in feature space. Other common misclassifications include LN_HIGH (35.0% for LWS, 27.8% for HS) and TICK_MID (19.2% for LWS, 15.0% for HS).

This performance gap highlights the challenge of open-set recognition in practice. While the distance-based approach works for some unknown types (MN, FH), others (HS, LWS) produce embeddings that lie within the acceptance regions of known classes. The wide variation suggests that geometric separability

in the 528-dimensional feature space depends heavily on the specific signal characteristics of each unknown jammer type.

Despite the limited unknown detection rate, if we consider the goal to be alerting operators to the presence of any jamming activity (rather than specifically identifying new jammers), the performance improves, reaching 63% of unknown, either by being correctly flagged as UNKNOWN or by being misclassified as one of the 16 known jammer types.

This means that even when OSR fails to recognize a signal as novel, the system still detects it as malicious activity in most cases. The distinction between "unknown jammer" and "known jammer" becomes less critical when the primary operational requirement is to warn users that an attack is occurring. From this perspective, the 37% that are misclassified as CLEAN represent the true failure cases, signals that goes through undetected.



Cut along the dashed line